

实验四：Simulink 的对离散系统和连续系统的建模与仿真

实验目的：

掌握离散系统和连续系统的建模方法，熟悉分层建模的方法。

实验内容：

1. This example implements population dynamics based on a nonlinear, discrete model. In the model, the population in a certain year, $p(k)$, depends proportionally on the population of the previous year, $p(k-1)$, multiplied by the reproduction rate, r . However, the resources are limited to L individuals, thereby creating negative effect on the population. The overall dynamics are given by the difference equation:
$$p[k]=p[k-1]+r*p[k-1]*(L-p[k-1])$$

For $L=1e6$ and $p(0)=1e5$, try
 $r=1.5e-6$ (system converges to a single value)
 $r=2.2e-6$ (system is periodic every 2 years)
 $r=2.5e-6$ (system is periodic every 4 years)
 $r=2.56e-6$ (system is periodic every 8 years)
a) Start with a Unit Delay block whose input is $p(k)$ and output is $p(k-1)$. Then construct the expression for $p(k)$, which is given in terms of the population. Run the system with a Stop time of 100 to observe the dynamics. Use a sample time of 1.
2. Computing a moving average is a commonly used technique to analyze trends in an otherwise wildly fluctuating(波动) set of time series data (e.g. stock prices or temperature of the atmosphere). The file `temp.xls` contains temperature data over several years. In this exercise, we will apply a moving average to this data to surmise long-term trends in the temperature of Earth's atmosphere.

First, you must load the data into the MATLAB workspace with the following command:

```
>> data=xlsread('temp.xls');
```

Then, build a model that computes a 6-point moving average. The equation is given by

$$y[k]=(x[k]+x[k-1]+x[k-2]+x[k-3]+x[k-4])/5$$

Start with a Unit Delay block whose input is $x(k)$ and output is $x(k-1)$. Connect the output with another Unit Delay block to obtain $x(k-2)$, and so on.

Additionally, you must

- Load the data(`data`) into the model using a from Workspace block.
 - Set the start time and stop time to be the first year(`data(1,1)`) and last year(`data(end,1)`) of the imported temperature data, respectively.
 - Set the sample time of the algorithm to 1 for every Unit Delay block, to match the period of the input data.
3. You are going to model a water tank. The equations that model the water level of the tank are

$$\dot{h} = \frac{1}{A * \rho} * (w_{in} - w_{out})$$

The derivative of the height is a function of the difference of the water flow in and out. h is the water level of the tank. A is the surface area of the bottom of the tank, and ρ is the density of the water. The flows in and out (w_{in} and w_{out}) are mass flows. The flow out is a function of water level h :

$$w_{out} = \left(\frac{1}{R}\right) \sqrt{\rho * g * h}$$

R is the resistance of the exit, and assume the exit is at the bottom of the tank.

- The combination of these two equations is a continuous nonlinear dynamical model of the water level in a tank. Build this model in Simulink.
- And create subsystem for this model.
- If the height of the tank is considered to be 2m, at what time does the tank fill completely?

Tips: outputs h —Height of the water in the tank

w_{out} ---water flow out of the tank

Parameters:

A ---Area of tank($4m^2$)

ρ ---Density of water($1000kg/m^3$)

w_{in} ---Water flow into the tank($100kg/s$)

R ---Resistance of the exit($5(kg.m)^{-1/2}$)

h_0 ---Initial height of the water in the tank(0 m)

g ---Gravitational constant($9.81m/s^2$)

h_{max} ---Height of tank($2m$)

States:

h ---height of the water in the tank

4. A free-falling ball is a good example of a system with continuous dynamics. The position of a ball in free fall is governed by Newton's Second Law:

$$m\ddot{x} = mg$$

Where m is the ball's mass, g is the gravitational acceleration, and x is the position. For simplicity air resistance is not considered here,

Since m is on both sides of the equation, the cancel out, and the equation simplifies to

$$\ddot{x} = g$$

- a) Build a Simlink model that emulates the dropping of a ball, Test the model with the following parameters and initial condition:

$$g = 9.81$$

$$x_0 = 0$$

$$\dot{x}_0 = 0$$

Run the simulation until a stop time of 10 seconds.

- b) Save the results into both the Matlab workspace and a MAT-file(using the to file block in source library).